

A Study on the Evolution of BMI in Belgian Employees

Project 3: Incomplete Data Analysis

Hasselt University

Submitted by:

Ambe Revelation Langsi
Harvey Nyerovwo Okotie
Pardon Masuka
Tiwonge Bindula
Tom Wellard Nangosyah

Presented to:

Prof. Dr. Geert Molenberghs
Prof. Dr. Geert Verbeke

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1 Abstract

Background: Body Mass Index (BMI) serves as a crucial metric for categorizing adults based on their heights or weights, providing insights into their size, whether classified as overweight or underweight. Several factors contribute to employees' BMI, with sex and age being notable influencers. Observations suggest that males generally exhibit higher BMI than females, and BMI tends to increase with age. The study focuses on understanding the evolution of BMI among Belgian employees over a span of 10 years, considering factors such as sex, age, and smoking status. The longitudinal nature of the study involves repeated BMI measurements over time, addressing the challenge of missing data to provide comprehensive insights.

Objectives: This study examined the BMI evolution of 4425 employees over 10 years, investigating the impact of sex and age. The baseline smoking status of the study participants was also accounted for.

Methodology: The analysis considered the incomplete nature of the data, employing three types of models: marginal model (Generalized Estimating Equations), Linear mixed model, and Generalized linear mixed model. To handle missing values, methods such as Direct Likelihood, Weighted Generalized Estimating Equations, and Multiple Imputation are applied, along with sensitivity analyses to account for the impact of different missing mechanisms.

Results: Consistent results were observed for both direct likelihood and multiple imputation concerning the continuous response. However, sensitivity analyses revealed that inference about the linear effect of sex is influenced by the change assumption from Missing at Random (MAR) to Missing Not at Random (MNAR). Similar observations are made for trichotomous outcomes, where Weighted Generalized Estimating Equations and Multiple Imputation yield comparable conclusions. Nevertheless, the linear effect of sex remained sensitive to the choice of methods.

Conclusions: The study found that age has a significant linear effect under both MAR and MNAR for both continuous and trichotomous outcomes. However, sex and baseline smoking exhibited non-significant effects on the trichotomous outcome.

Key words: Missingness, Direct Likelihood, Multiple Imputation, Weighted Generalized Estimating Equations, Sensitivity Analysis.

2 Introduction

In health assessment, Body Mass Index (BMI) stands as a widely recognized metric, providing a numerical estimation based on an individual's weight and height. While it may not directly measure body fat, BMI has consistently demonstrated a robust correlation with more intricate methods for body fat assessment, such as underwater weighing and dual energy x-ray absorptiometry (DXA). Its appeal lies in its cost-effectiveness and simplicity, making it a valuable initial screening tool to classify individuals into weight categories associated with potential health risks.

The World Health Organization (WHO) categorises BMI as follows: Below 18.5 (underweight), 18.5 – 24.9 (normal), 25.0 – 29.9 (overweight), and 30.0 and above (obese). It is essential to understand the evolution of BMI and the potential links to age, gender, and smoking behavior amongst people. This study analyses longitudinal BMI measurements, focusing on a sample of Belgian employees who underwent annual medical examinations. These employees' BMI values were recorded at a maximum of 10 different time points, forming the basis for our exploration and insights.

3 Research objective and Research Question

The primary objective is to study the BMI evolution of a sample of the Belgian employee population. Specifically, we aim to address the following key questions:

- **BMI Evolution Over Time:** How does BMI change over the course of the study, and are there any trends or patterns in the data?
- **Gender Differences:** Does BMI evolution vary significantly between male and female employees?
- **Age Influence:** Does the initial age of the employee at the start of the study affect the BMI evolution?

4 Data Description

The data in this study are longitudinal measurements of the body mass index (BMI) for a sample of 4425 Belgian employees who are invited for yearly medical examinations. The dataset comprises information collected from these individuals at a maximum of 10 time points. The variables included in the data are subject id, time described as the time since entry in the study (years), BMI refers to the Body Mass Index measured in kg/m^2 , fage is the age of the employee at the first measurement (years), sex refers to the gender of the employee (0: female, 1: male) and smoking refers to the smoking status at the time of measurement (0: non-smoker, 1: smoker).

We will be using two versions of the BMI: A continuous version and a trichotomized version of the BMI as the outcome variables of interest, with the other variables fage, time, sex, smoking as possible covariates. BMICAT was defined as:

$$\text{BMICAT} = \begin{cases} 1 & \text{if BMI} \leq 25, \\ 2 & \text{if } 25 < \text{BMI} \leq 30, \\ 3 & \text{if BMI} > 30 \end{cases}$$

The measurements were taken at fixed and equally spaced time points for all subjects. However, every subject did not have the same number of measurements. Therefore, the analysis takes into account the unbalanced nature of the data specifically the time intervals and the measurements.

5 Methodology

5.1 Exploratory Data Analysis

The exploration of missingness aimed to provide a comprehensive understanding of the missingness patterns within the dataset. This involved an examination of the proportion of completers, as well as an analysis of monotone and non-monotone missingness patterns. Additionally, factors influencing missingness were investigated to enhance insights into this crucial aspect of the dataset.

5.2 Missing Data Mechanisms

Missingness often happens whenever subjects originally recruited fail to participate in one or more measurement periods, or when measurements are lost, or otherwise unavailable (e.g. when the subject dies). When a subject leaves the study prematurely, the missing data pattern is termed as dropout/monotone missing, while intermittent participation of the subject would result to non-monotone missingness.

Analysis of missing data is usually based on assumptions about the process that causes the missingness. [Little and Rubin \(2002\)](#) classified the missing data mechanism into three categories; Missing Completely At Random (MCAR), Missing At Random (MAR), and Missing Not At Random (MNAR). MCAR describes a process in which the missingness does not depend on either the observed or missing values. On the other hand, the MAR is when the missingness conditionally depends only on the observed data whereas it is MNAR if the missingness depends on both the observed and unobserved data

It is evident from literature ([Mazumdar et al., 2007](#); [Lee, 2007](#); and [EMEA/CPMP, 2001](#)) that the quality of inference made when data are incomplete critically depends on the assumption made about the missing mechanism. Thus, recognising the type of this mechanism is vital since it would help clarify under what conditions it could be possible to ignore the missingness and yet have valid inference. However, it might be challenging to attribute the missingness to any of the three mechanisms (MCAR, MAR, or MNAR) since missing information is not observed, hence, it is not possible to know how the complete sample looks like. As mentioned by [Molenberghs and Verbeke \(2005\)](#), sensitivity analysis would be the most sensible action to perform whenever assumptions are made about the missing mechanism.

The traditional methods for handling missing data are based on simple methods such as Complete-Case analysis (CC) and Last Observation Carried Forward (LOCF). However, these methods are based on strong assumptions about the missing values. For example, CCA is valid only under

MCAR and involves information loss. Also, the validity of LOCF is based on strong and unrealistic assumptions that the subjects' measurement remains the same as previous (Molenberghs and Kenward, 2007). Several authors (Mazumdar et al., 2007; Molenberghs and Verbeke, 2005; and, Albert and Wu, 2004) have discussed other inferential procedures including likelihood based and frequentist methods, for the analysis of longitudinal data with missing values.

5.3 Models

In this report, Linear Mixed Model (LMM), Generalized Linear Mixed Model (GLMM) and Marginal Model (GEE) were used for modelling the continuous and trichotomous responses respectively, while disregarding the presence of missingness. Let $\mathbf{Y}_i$ and $\mathbf{Z}_i$ denote the continuous and trichotomous response taken for i ($i = 1, \dots, 4425$) at fixed set of time points $t_0, \dots, t_9$. The mixed effects models are specified below:

$$\begin{aligned} \eta_{ij} = & \beta_0 + b_{1i} + \beta_1 \text{fage}_i + \beta_2 \text{sex}_i \\ & + \beta_3 \text{baseline_smoking}_i + \beta_4 \text{time}_{ij} + (b_{2i} + \beta_5 \text{fage}_i + \beta_6 \text{sex}_i) \times \text{time}_{ij} + \epsilon_{ij} \end{aligned} \quad (1)$$

where $\mathbf{b}_i = (b_{1i}, b_{2i})'$ and $\mathbf{b}_i \sim \mathcal{N}(0, \mathbf{D})$ assumed to be independent of $\epsilon_{ij} \sim \mathcal{N}(0, \mathbf{\Sigma})$. For the continuous response, the LMM model is given by:

$$Y_{ij} = \eta_{ij} \quad (2)$$

While for the trichotomous response, assuming $Z_{ij} | \mathbf{b}_i \sim \text{Multinomial}(\boldsymbol{\pi}_{ij})$, the GLMM model is given by:

$$\begin{aligned} \text{Logit}(\boldsymbol{\pi}_{ij}) = & \beta_0 + b_{1i} + \beta_1 \text{fage}_i + \beta_2 \text{sex}_i \\ & + \beta_3 \text{baseline_smoking}_i + \beta_4 \text{time}_{ij} \\ & + (b_{2i} + \beta_5 \text{fage}_i + \beta_6 \text{sex}_i) \times \text{time}_{ij} + \epsilon_{ij} \end{aligned} \quad (3)$$

For the trinomial outcome in this study where the multinomial distribution and cumulative logit link function has the same formulation as proportional odds logistic regression. The initial GEE model formulation is given as;

$$\begin{aligned} \text{Logit}[\boldsymbol{\pi}_{ij}] = & \beta_0 + \beta_1 \text{fage}_i + \beta_2 \text{sex}_i + \beta_3 \text{baseline_smoking}_i \\ & + \beta_4 \text{time}_{ij} + (\beta_5 \text{fage}_i + \beta_6 \text{sex}_i + \beta_7 \text{baseline_smoking}_i) \times \text{time}_{ij} \end{aligned} \quad (4)$$

where; $\boldsymbol{\pi}_{ij} = P(Y_{ij} \leq k | t_{ij}, X_i)$

The two models specified above (LMM and GLMM) assume ignorable missingness, which is valid only under Missing Completely at Random (MCAR) and Missing at Random (MAR) conditions (see Molenberghs and Verbeke, 2005; Hedeker and Gibbons, 2006; and Sotto, 2009). In the case of non-ignorable missingness, techniques that can handle this missingness need to be employed. Therefore, it was important to investigate the missing mechanism exhibited by the BMI data.

This was done by modeling the probability of dropping out given the previous measurement and other covariates in the dataset. Let $\mathbf{R}_i = (R_{i0}, \dots, R_{i9})'$, where $R_{it} = 1$ if Y_{it} is observed and 0 otherwise. Also, let $D_i = 1$ if drop-out occurs at that time and 0 otherwise. Then, the probability of dropping out at a particular time t can be modeled as:

$$\text{logit}(P(D_i = 1)) = \gamma_0 + \gamma_1 \text{fage}_i + \gamma_2 \text{sex}_i + \gamma_3 \text{baseline_smoking}_i + \gamma_4 \text{prev}_i \quad (5)$$

where prev_i is the bmi for individual i at the previous time point. If $\gamma_4 = 0$, then this signifies the possibility of MAR, while if $\gamma_4 \neq 0$, it signifies the possibility of MNAR (Hedeker and Gibbons, 2006). When the missing mechanism is found to be MAR, Generalized Estimating Equations (GEE) provide a valid and robust alternative to the full likelihood-based procedures specified above (Sotto, 2009). This method is especially useful when the data are non-Gaussian and interest is restricted to the mean parameters. However, when the missing mechanism is no longer MAR, a wrongly specified working correlation matrix may lead to additional bias. Hence, methods such as weighted GEE (WGEE) and multiple-imputation GEE (MI-GEE) for MAR data have been proposed (see Molenberghs and Verbeke, 2005).

The general idea behind WGEE is to use the inverse probability as weights derived from a dropout model. MI-GEE, on the other hand, employs imputation procedure to replace the missing values (monotone and non-monotone) before analysis is conducted. Under this approach, the completed datasets (generated under M imputations) are separately analyzed and subsequently pooled for final inference (Molenberghs and Verbeke, 2005). Although both WGEE and MI-GEE can handle missing data under MAR, Sotto (2009) showed that WGEE is sensitive to misspecification in either, the dropout or measurement model, while MI-GEE is fairly robust to misspecification in either the imputation or measurement model, thus yielding more precise estimates. To deal with missingness therefore, MI-GEE was used to obtain marginal inference on the trichotomous response. In addition, the multiple imputation concept was extended to the likelihood-based models (MI-GLMM and MI-LMM). Note that, although all the methods discussed so far encompass both MCAR and MAR, it is still inappropriate to rule out the notion of a more general missingness mechanism (MNAR). And since it is not known what the true underlying mechanism driving the missingness is, more confidence in the conclusions can be gained by means of sensitivity analysis (Molenberghs and Verbeke, 2005).

5.4 Sensitivity Analysis

In the sensitivity analysis pattern mixture models were employed to assess the impact of different missing mechanisms. These included Complete Case Missing Values (CCMV) - includes only cases with complete information assuming missingness is completely unrelated to observed or unobserved data, here if results align with those under MAR, this supports the notion that missing values are MCAR, Neighbouring Case Missing Values (NCMV) - includes cases with missing values as a distinct group, allowing for the possibility that missingness is related to observed values but not to unobserved values, for this group significant effects may indicate departure from MAR, suggesting MNAR and All Cases Missing Values (ACMV), includes all cases considering missingness related to both observed and unobserved values, here a departure from MAR in ACMV indicates a systematic pattern of missingness. These models explicitly incorporate missing data patterns into the analysis allowing for exploration of different relationships

between missingness and observed/unobserved variables. Given we had varying missing data mechanisms in our dataset, understanding the variations and how they influence the parameter estimates was essential for robust statistical inference.

5.5 Software

We carried out our analyses using both R software, version 4.2.1. and SAS Software, version 9.4 and a significance level(α) for decision-making was set at 0.05.

6 Results

6.1 Exploratory Data Analysis

We shall present an exploratory analysis of the continuous and trichotomous outcomes of BMI as a tool to provide insight to the modelling procedure. Starting with the continuous outcome, the overall average evolution shows a slight increase over time with a slightly linear trend in BMI over time (see figure 1a). The average evolution by gender shows differences between males and females with males having higher BMI values compared to females, see figure 1b.

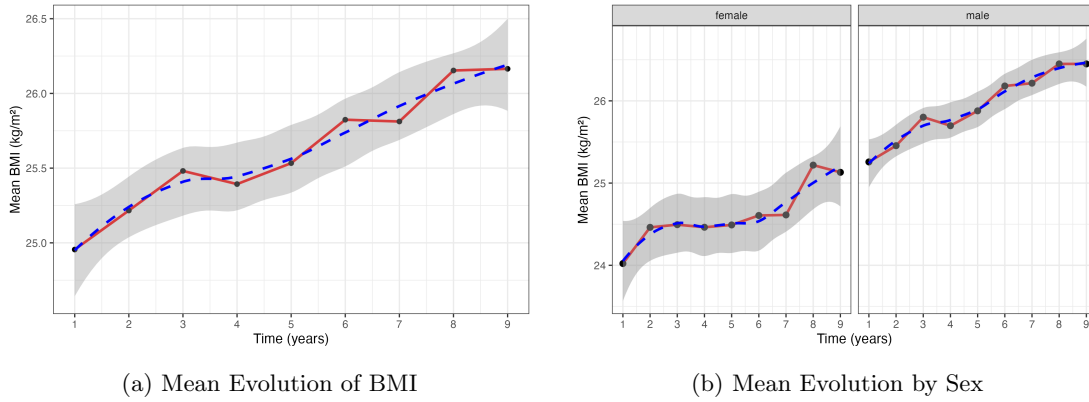


Figure 3 illustrates how subjects BMI evolved overtime. The plot suggested the existence of large between and less within subject variability. The implication of this on the modelling, suggested the use of random intercepts model to explain the between subject variability in the data. The figure also shows that some subjects dropped out early from the study.

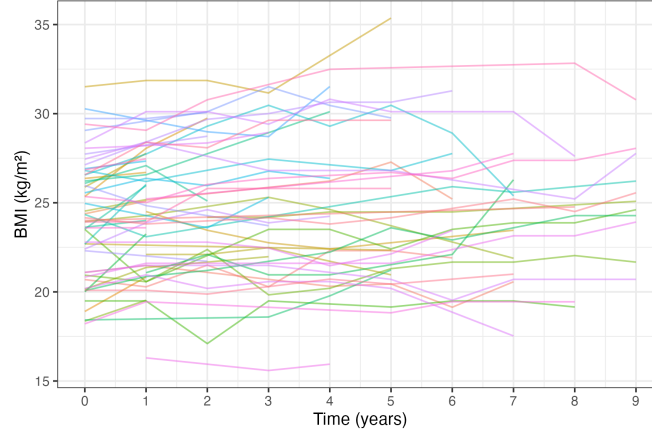


Figure 2: Individual Profiles plot of 100 selected ID's

We did similar exploration for the trichotomous outcome, where the overall average evolution for the subjects across different BMI categories was plotted. We sampled 20 ids and plotted their BMI over time, where each line corresponds to a unique subject, with lines colored in red showing the transitions between the BMI categories. The thickness of the line denotes instances where the individual transitions from one BMI category to another.

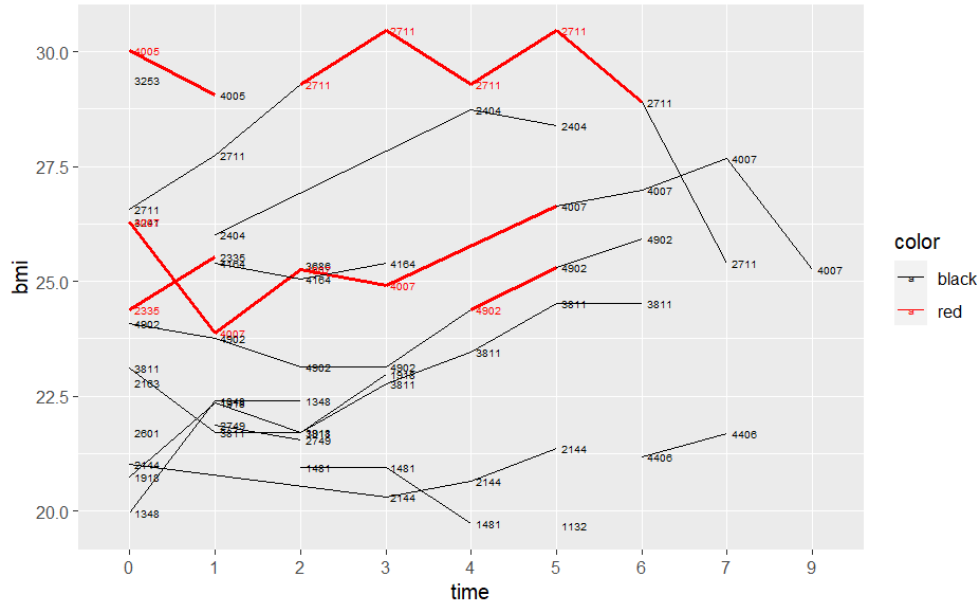


Figure 3: BMI Evolution by category for 20 selected ID's

6.1.1 Exploring Missingness

As observed earlier in figure 3, some subjects dropped out early from the study. This gave us a narrow picture of the missing pattern in the data. In table 1 we see a complete picture of

missing patterns in the data; around 55.6% of the subjects dropped out early, while 42.3% had intermittent missing values. A number of patterns were identified in the data, but we show the most prevalent.

Table 1: Missing Patterns

0	1	2	3	4	5	6	7	8	9	Frequency	Percentage
Completers											
O	O	O	O	O	O	O	O	O	O	92	2.07%
Dropouts											
O	M	M	M	M	M	M	M	M	M	1417	32.02%
O	O	M	M	M	M	M	M	M	M	404	9.12%
O	O	O	M	M	M	M	M	M	M	228	5.15%
O	O	O	O	M	M	M	M	M	M	158	5.51%
O	O	O	O	O	O	O	O	O	M	36	0.81%
O	O	O	O	O	M	M	M	M	M	86	1.94%
.	.	.	.	.	.	.	.	.	.	.	.
O	O	O	O	O	O	M	M	M	M	59	1.33%
Non-Montone Missingness											
O	M	O	M	M	M	M	M	M	M	122	2.75%
M	O	M	M	M	M	M	M	M	M	81	1.83%
O	M	O	O	M	M	M	M	M	M	77	1.74%
O	O	M	O	M	M	M	M	M	M	45	1.01%
O	M	M	O	O	O	M	M	M	M	33	0.74%
O	M	O	O	O	M	M	M	M	M	32	0.72%
.	.	.	.	.	.	.	.	.	.	.	.
O	O	O	O	O	O	O	O	M	O	24	0.54%

To verify whether dropout was related to known factors such as age, sex, baseline smoking status as well as the previous outcome, a logistic model was fitted, while ignoring the non monotone missingness. Parameter estimates and standard errors are presented in Table 2. All the effects such as; sex, baseline smoking and previous outcome were found significant, implying that dropout mechanism was dependent of the known covariates in the model. Further, suggesting that there is MAR and any form of bias introduced by the missing data will be accounted for through the observed variables.

Table 2: Parameter Estimates for the Dropout Model

Effect	Parameter	Estimate (s.e)	p-value
Intercept	γ_0	−28.2304(0.1044)	< 0.0001
Age	γ_1	−0.0159(0.0020)	< 0.0001
Female	γ_2	25.9959(< 0.0001)	< 0.0001
Baseline Smoking	γ_3	0.2977(0.0352)	< 0.0001
Previous Outcome	γ_4	0.0104(0.0038)	0.0068

6.2 Linear Mixed Model

6.2.1 Modelling the Continuous Outcome ignoring the missing mechanism

A Linear Mixed Model was fitted for the continuous BMI outcomes while ignoring the missingness mechanism in the data. Parameter estimates and empirical standard errors for the obtained values are presented in Table 3. The model shows that an almost linear trend explains the average evolution in the BMI. In addition, the model also showed a significant effect for age, sex, baseline smoking, time and the interaction between age and time. We can say that the average evolution of BMI amongst Belgium employees can be explained by these significant variables.

Table 3: Parameter estimates (empirical standard errors) for the LMM for BMI ignoring the missing mechanism

Effect	Parameter	Estimates (s.e)
Intercept	β_0	20.901(0.2006)*
Age	β_1	0.115(0.0054)*
Female	β_3	−0.787(0.133)*
Baseline Smoking	β_2	0.446(0.117)*
Time	β_4	0.341(0.0545)*
Time*Age	β_6	−0.005(0.0014)*
Time*Female	β_7	−0.0625(0.032)
Time*Baseline Smoking	β_8	0.0059(0.0149)
Variance (b_{1i})	d_{11}	12.658
Variance (b_{2i})	d_{22}	0.0545
Covariance (b_{1i}, b_{2i})	d_{12}	−0.0225
Residual variance (ϵ_{ij})	σ^2	0.989

6.2.2 Modelling the continuous outcome correcting for missing mechanism

The likelihood-based procedures used to analyze the continuous outcome, are both valid under the MAR and MCAR assumption. This is so since under the likelihood framework, missing data generated under MAR or MCAR can be considered ignorable. In addition, methods based on the MAR assumption are valid if data are MCAR or MAR, but MCAR methods are valid only if data are MCAR (Mallinckrodt et al., 2003). With this in mind, the LMM was refitted using direct likelihood (complete cases) and multiple imputation as a way to deal with the missing

mechanism. Parameter estimates and robust standard errors for the model analyses are presented in Table 4

Comparing the parameter estimates, it's observed that applying direct likelihood analyses and multiple imputation resulted in almost similar estimates with the direct likelihood slightly underestimated for the model parameters.

Table 4: Parameter estimates for LMM on Direct Likelihood (complete cases) and Multiple Imputation

Effect	Direct Likelihood			MI		
	Estimate	SE	P-value	Estimate	SE	P-value
Intercept	20.901	0.199	<.0001	21.130	0.2111	<.0001
Age	0.115	0.0054	<.0001	0.111	0.0059	<.0001
Baseline Smoking	0.470	0.114	<.0001	0.416	0.121	0.0006
Female	-0.800	0.132	<.0001	-0.783	0.131	<.0001
Time	0.341	0.0273	<.0001	0.332	0.0452	<.0001
Time*Age	-0.0055	0.0008	<.0001	-0.0054	0.0015	0.003
Time*Female	-0.0217	0.0169	0.199	-0.0035	0.0155	0.820
Time*Baseline Smoking	0.0059	0.0149	0.692	0.0056	0.0121	0.644

6.3 Generalized Linear Mixed Model

We fitted a generalised linear mixed model (GLMM) under the assumption of an MAR missing data mechanism. First, multiple imputation was done using the Markov Chain Monte Carlo (MCMC) simulation model, which is suitable for data with monotone and non-monotone missingness. Five (5) imputations were done, which is within the 3 - 10 recommended by [Rubin \(1978\)](#). A non-adaptive Gaussian quadrature, 40 quadrature points GLMM model was then fitted using the imputed data. The results from the multiple imputed datasets were combined into a single inference and the parameter estimates from the resulting model are shown in table 5.

Table 5: Parameter estimates for GLMM on Multiple Imputation

Parameter	Estimate	Std Error	P-value
Intercept 1	9.246	0.037	< .0001
Intercept 2	17.467	0.189	< .0001
Time	0.639	0.0620	< .0001
Female	1.986	0.254	< .0001
Age	0.218	0.010	< .0001
Baseline Smoking	-0.994	0.216	< .0001
Time*Female	-0.019	0.032	0.569
Time*Age	-0.010	0.00137	< .0001

Conditioning on the random effects, age and female were significant and associated with the BMI category while baseline smoking status was not. The interaction between time and age

was significant ($p < 0.0001$), suggesting that an increase in time decreases the log odds of being in a lower BMI category for older subjects compared to younger subjects. The time and sex interaction was not statistically significant ($p = 0.569$).

The random intercept and random slope in table 6 show significant variability across subjects, as indicated by the significant estimates for d_{11} and d_{22} . There's also a significant negative covariance between the intercept and slope (d_{12}), suggesting that subjects with higher intercepts tend to have less steep slopes, or vice versa. These results highlight individual differences in the evolution over time. Notably, the participants' starting BMI categories were very variable.

Table 6: Parameter estimates for Random Effects

Parameter	Estimate	Standard Error	Pr > t
d_{11}	40.958	1.560	< .0001
d_{12}	-0.291	0.0915	0.0015
d_{22}	0.129	0.0073	< .0001

6.4 Generalized Estimating Equations

The analysis was further extended to GEE framework, given the standard GEE is only valid under MCAR, two alternative models were explored to accommodate the MAR. Weighted GEE and Multiple Imputation GEE were fitted to the data and the parameter estimates and standard errors are presented in table 7. To address the non-monotone missingness, a macro dropout function was employed to create weights for WGEE, accounting for the intermittent missingness patterns in the data allowing for accurate analysis. For inference the empirical estimates were used to calculate the probability of an individual belonging to a given BMI category. The methodology used follows from [Robins et al. \(1995\)](#), where weights were assigned to each subject post-monotonising observed intermittent missingness patterns.

Table 7: Results of Weighted GEE versus Multiple Imputation GEE

Effects	WGEE			MI GEE		
	Estimates	SE	P-value	Estimates	SE	P-value
Intercept 1	2.447	0.198	<.0001	2.181	0.105	<.0001
Intercept 2	4.077	0.188	<.0001	4.2580	0.117	<.0001
Time	0.007	0.042	0.8701	-0.175	0.011	<.0001
Age	-0.054	0.005	<.0001	-0.056	0.003	<.0001
Female	0.389	0.133	0.0065	0.440	0.072	<.0001
Baseline Smoking	-0.296	0.106	0.0059	-0.235	0.061	<.0001
Time*Baseline smoking	0.009	0.018	0.6015	0.001	0.007	0.8588
Time*Age	0.005	0.001	<.0001	0.003	0.00027	<.0001
Time*Female	0.041	0.040	0.3287	-0.003	0.007	0.7223

For the standard GEE, 10 imputations were performed for the missing values in each dataset before the pooling of results to get the final estimates. The findings revealed significant effects

of age in both WGEE and MI GEE models (p-value = 0.001). Similarly, sex emerged as a significant factor in both models (p-value = 0.0065 for WGEE and 0.0001 for MI GEE). Baseline smoking also exhibited significance in both models (p-value = 0.0059 for WGEE and 0.0001 for MI GEE). For the time variable, significance was observed under standard GEE after multiple imputations (p-value = 0.001). The interaction between time and age demonstrated significance in both models (p-value = 0.001), suggesting that an increase in time increases the probability of being in a low BMI category for older subjects compared to younger subjects.

6.5 Comparison of results

In our analysis we looked at the exploratory data analysis conducted in the first and second assignment to compare the results with our current fitted models. Our examination looked at individual profiles, mean structure and the variance function for comparison.

It was seen in the EDA (figure 1b) that the BMI evolution for males tends to be higher on average compared to females, the mean structure showed a gradual increase in BMI over the years with a slight deviation in linearity while the variance function was estimated to change slightly overtime. When comparing the results with the fitted models, the following observations become apparent:

- The results of the D matrix in table 6 indicate substantial variability among employees at the beginning of the study regarding their initial BMI values ($d_{11} = 40.958$, $d_{22}=0.129$). It suggests individuals with higher initial BMI values tend to exhibit a slower rate of increase in BMI compared to those with lower initial BMI values ($d_{12} = -0.291$).
- Looking at the variance function over time, looking at the random intercepts, we see that there's high variability in the intercepts compared to the slopes ($b_{1i}=12.658$ vs $b_{2i}=0.0545$); this is consistent with what was seen in the individual plots but also with the variance function, which had small variability over time.
- The mean structure exhibited a subtle deviation from linearity in the evolution of BMI over time. Despite this, the significance test for the quadratic time effect rendered it insignificant. For our current models, we opted for a linear evolution of BMI, illustrating this linear evolution over time in both instances.
- The difference in BMI evolution between males and females, as observed in the EDA, aligns with the estimates for the sex variable in our models. The negative coefficient suggests that females exhibit a lower evolution in BMI compared to males, specifically indicating that over time, the BMI evolution for females decreases by 0.0625 kg/m^2 in comparison to male employees.

Considering the observations, differences between the EDA and the model results may arise due to variations in the modeling assumptions and methods used for analysis, for example, the imputation methods and the missingness criteria considered. In the context of our models, the differences appear relatively minor and may not cause a significant problem.

6.6 Sensitivity Analysis

Based on the findings presented in Table 8, it is evident that there is an indication against the assumption of Missing Completely At Random (MCAR). While full-likelihood based methods are generally valid under the Missing At Random (MAR) assumption, there are situations where MAR might be violated. In view of this, [Molenberghs and Verbeke \(2005\)](#) advise to always perform a formal sensitivity analysis to identify any discrepancies in the outcomes of models under different missing mechanisms.

The results for the Gaussian outcome, as detailed in table 4, involved an assessment of deviations in the assumption of MAR and Missing Not At Random (MNAR) methods for the multiple imputed data. If the conclusions drawn from this scenario differ from those under the MAR assumption, it raises concerns about the validity of the MAR assumption. Under the MNAR assumption, missing values were imputed using the pattern-mixture models approach, considering the NCMV ACMV, and CCMV identifying restrictions.

The results presented in Table 3 revealed that the parameter estimates and standard errors obtained through these two methods were very identical to those from multiple imputations in Table 4. Regarding the significance of the covariates, the majority of them showed consistent significance in the MI-LMM model and the three models used for sensitivity analysis. This suggests that there is no substantial deviation from the MAR assumption.

Table 8: Sensitivity analysis estimates for LMM

Effects	CCMV (SE)	NCMV(SE)	ACMV(SE)
Intercept	20.846(0.191)*	20.921(0.194)*	20.927(0.192)*
Time	0.341(0.016)*	0.434(0.017)*	0.331(0.0136)*
Age	0.118(0.005)*	0.117(0.005)*	0.116(0.005)*
Female	-0.919(0.129)*	-0.728(0.129)*	-0.806(0.129)*
Baseline smoking	0.462(0.113)*	0.403(0.115)*	0.434(0.114)*
Time*Baseline smoking	0.0037(0.009)	0.018(0.012)	0.007(0.012)
Time*Age	-0.0055(0.0005)*	-0.009(0.0005)*	-0.0054(0.0004)*
Time*Female	-0.031(0.012)*	-0.056(0.014)*	-0.015(0.013)

7 Discussion and Conclusion

In this research, we explored the dynamics of Body Mass Index (BMI) over a span of time and explored its correlation with the age and gender of subjects, acknowledging that certain patients have missing measurements at different time points. Missing data is a common occurrence in longitudinal settings, and our study’s data was no exception. Our dropout investigation revealed that only 2.07% were completers, with the majority exhibiting monotone missingness (55.6%), while 42.3% had non-monotone missingness at various times during the measurement period.

To gain insights into factors influencing missingness, we employed an ordinary logistic regression conditioned on the previous BMI outcome and other covariates; age, sex and baseline smoking status. Interestingly, the findings indicated that the previous BMI outcome and the other covariates were statistically significant, questioning the assumption of Missing Completely At

Random (MCAR). Analysis from the previous assignments showed that the model with both the intercept and linear time slope was the most parsimonious.

We fitted three different types of models in order to answer the above research questions: Linear Mixed Model (LMM), Generalised Estimating Equations (GEE) and Generalised linear mixed models (GLMM). Each of these models accounted for the missingness mechanism. The LMM model using complete cases showed significant bmi evolution over time by both age and sex. The LMM model with direct likelihood and multiple imputations had similar estimates and the BMI evolution differed by age and not sex. The GLMM model with multiple imputation showed significant BMI evolution over time by age but not by sex. The weighted GEE and multiple imputations GEE models had very similar coefficient estimates. Both of these models also showed significant BMI evolution for age but not for sex.

Following recommendations from [Molenberghs and Verbeke \(2005\)](#), a comprehensive sensitivity analysis was carried out to examine departures from the assumed missingness mechanisms. For the multiple imputation data in the Gaussian model, pattern mixture models were employed, and the results indicated no substantial evidence against the assumption of a Missing At Random (MAR) mechanism. Since missing not at random could not be ruled out, we did a sensitivity analysis using NCMV, ACMV, and CCMV, which returned similar estimates and standard errors.

In conclusion, our analysis supported the suitability of Missing At Random (MAR) as the missing mechanism in this dataset. Overall, the BMI tends to change over time, and this progression is contingent on the baseline age of the subjects. Furthermore, the average evolution of BMI over time did not significantly differ between males and females in this study.

8 References

References

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9 Codes

```
**R codes**

% Individual profile plot

# Load necessary libraries
library(ggplot2)
library(dplyr)

# Assuming your code up to this point is executed
selected_ids <- sample(unique(lda_baseline_smoking$id), 20,
  replace = FALSE)

# Create a new data frame with all rows for the selected IDs
sampled_data <- lda_baseline_smoking %>%
  filter(id %in% selected_ids)

# If you want to reset the row names or create a new index
sampled_data <- sampled_data %>% rownames_to_column()

# Define categories based on BMI
category_thresholds <- c(25, 30)

# Create a new column indicating the starting and ending
  categories
sampled_data <- sampled_data %>%
  group_by(id) %>%
  mutate(start_category = ifelse(bmi < category_thresholds[1], "
    <25", ifelse(bmi < category_thresholds[2], "25-30", ">=30")),
    end_category = ifelse(lead(bmi, default = last(bmi)) <
      category_thresholds[1], "<25", ifelse(lead(bmi, default = last(
        bmi)) < category_thresholds[2], "25-30", ">=30")),
    color = ifelse(start_category != end_category, "red", "
      black"))

# Plot the data with colored lines and text
ggplot(sampled_data, aes(x = as.factor(time), y = bmi, group = id,
  color = color)) +
  geom_line(size = ifelse(sampled_data$color == "red", 1, 0.5)) +
  geom_text(aes(label = id, color = color), hjust = -0.2, vjust =
    0.5, size = 2) +
  scale_color_manual(values = c("red" = "red", "black" = "black"))
+
```

```
labs(x = "time", y = "bmi")

**SAS codes**

Direct Likelihood LMM
/* Model gaussian serial correlation */
proc mixed data = Bmilda_bmicat1 method = reml ;
class id sex baseline_smoking time ;
model bmi = fage baseline_smoking sex timecont fage * timecont
sex *
timecont baseline_smoking * timecont
/ solution chisq ;
random intercept timecont / type = un subject = id g gcorr v vcorr
solution ;
repeated time / type = sp ( gau ) ( time ) local subject = id r
rcorr ;
run ; quit,

**Multiple imputation LMM**
/* MULTIPLE IMPUTATION*/
/* convert the data into wide format*/
proc sort data= Bmilda_bmicat;
by id time; run;

proc transpose data= Bmilda_bmicat out = bmi_mistep1;
var bmi;
id time;
by id fage sex baseline_smoking;
run;

/* not monotone imputation( default mcmc from page 633 in the
notes)*/
proc sort data=bmi_mistep1; by fage; run;

proc mi data=bmi_mistep1 seed=2023 simple nimpute=10 round=0.1 out
=bmi_imput;
var baseline_smoking _0 _1 _2 _3 _4 _5 _6 _7 _8 _9;
by fage;
run;
/* transposing back to the original format*/

data bmi_long_im;
set bmi_imput;
array times {*} _0-_9;
```

```
do i = 1 to dim(times);
    time = i - 1;
    bmi = times[i];
    output;
end;
drop i _0 -_9;
run;
/*sort by id*/
proc sort data=bmi_long_im;
by id;
run;

/* creating timecont and timecont2*/

data bmi_long_im;
set bmi_long_im;
timecont = time;
run;

data bmi_long_smokecat; /* Replace 'your_dataset' with the actual
dataset name */
set bmi_long_im;

/* Create baseline_smokingcat variable based on the condition
*/
if baseline_smoking <= 0 then
    baseline_smokingcat = 0;
else
    baseline_smokingcat = 1;

/* Optional: Drop the original baseline_smoking variable if
needed */
drop baseline_smoking;
run;

/* sort data by imputation*/

proc sort data=bmi_long_smokecat; by _imputation_; run;

/* fitting lmm with mi data*/
proc mixed data =bmi_long_smokecat METHOD=reml;
class id sex baseline_smokingcat time ;
by _Imputation_;
model bmi = fage baseline_smokingcat sex timecont fage * timecont
sex *
```

```
timecont / solution chisq covb;
random intercept timecont / type = un subject = id g gcorr v
      vcorr
solution ;
repeated time / type = sp ( gau ) ( time ) local subject = id r
      rcorr ;
ods output solutionF=work.nlparms CovB=work.nlcovb;

run ; quit;

/*Pool estimates for inference*/
proc mianalyze parms=work.nlparms;
class baseline_smokingcat sex;
modeleffects Intercept fage baseline_smokingcat sex timecont fage
      * timecont sex *
timecont ;
ods output ParameterEstimates = combined_parms;
run; quit;

% GLMM model

proc nlmixed data=Tmp1.final_data noad qpoints=40;
  title 'BMI, PROC_NLMIXED, ordinal, non_adaptive_Gaussian_
    quadrature, 40';
  by _imputation_;
  parms int1=2.283 int2=4.340 beta1=-0.188 beta2=0.511 beta3
    =-0.046 beta4=-0.237 beta5=0.028 beta6=0.002 d11=8.2003 d12
    =0.2871 d22=0.0109;
  eta = beta1*timecont + beta2*sex + beta3*fage + beta4*baseline_
    smoking_cat + beta5*timecont*sex + beta6*timecont*fage + b;
  if bmicat=1 then z = 1/(1+exp(-(int1-eta)));
  else if bmicat=2 then z = 1/(1+exp(-(int2-eta))) - 1/(1+exp(-(
    int1-eta)));
  else z = 1 - 1/(1+exp(-(int2-eta)));
  if z > 1e-8 then ll = log(z);
  else ll = -1e100;
  model bmicat ~ general(ll);
random b ~ normal(0,d*d) subject=id;
ods output      ParameterEstimates=Tmp1.nlparms
                  CovMatParmEst=Tmp1.nlcovb
                  AdditionalEstimates=Tmp1.nlparmsa
                  CovMatAddEst=Tmp1.nlcovba;

run;

#####
```

```
PROC IMPORT OUT= datasets.Bmilda_wide
            DATAFILE= 'C:\Users\harve\Downloads\wide_bmi.csv'
            DBMS=DLM REPLACE;
            DELIMITER=',';
            GETNAMES=YES;
            DATAROW=2;
RUN;
proc sort data=datasets.Bmilda_wide;
  by sex;
run;
proc mi data=datasets.Bmilda_wide seed=54321 simple nimpute=10
  round= 0.1 out=datasets.bmi_impute;
title 'Monotone multiple imputation';
by sex;
var fage baseline_smoker t0 t1 t2 t3 t4 t5 t6 t7 t8 t9;
mcmc impute = monotone;
EM maxiter = 500;
run;
proc sort data=datasets.bmi_impute;
  by _imputation_ id fage baseline_smoker sex;
run;
proc sort data=datasets.bmi_impute;
  by sex;
run;
PROC MI DATA=datasets.bmi_impute SEED=54321 OUT=datasets.bmi_
  impute
SIMPLE NIMPUTE=1 ROUND=0.1 MINIMUM=0;
VAR fage baseline_smoker t0 t1 t2 t3 t4 t5 t6 t7 t8 t9;
BY sex; RUN;

/*Transpose from wide to long format*/
DATA datasets.bmi_longMI;
SET datasets.bmi_impute;
time=0;Bmi=t0;OUTPUT;
time=1;Bmi=t1;OUTPUT;
time=2;Bmi=t2;OUTPUT;
time=3;Bmi=t3;OUTPUT;
time=4;Bmi=t4;OUTPUT;
time=5;Bmi=t5;OUTPUT;
time=6;Bmi=t6;OUTPUT;
time=7;Bmi=t7;OUTPUT;
time=8;Bmi=t8;OUTPUT;
time=9;Bmi=t9;OUTPUT;
DROP t0-t9;
RUN;
```

```
%dropout(data=datasets.bmi_longMI, id=id, time=time, response=bmi, out
          =datasets.bmi_mono);

proc genmod data=datasets.bmi_mono descending;
class sex baseline_smoker;
model dropout = fage sex baseline_smoker prev / dist=b pred;
ods output obstats=pred;
run;

data pred;
set pred;
keep observation pred;
run;
data datasets.bmi_mono;
merge pred datasets.bmi_mono;
run;

%dropwgt(data=datasets.bmi_mono, id=id, time=time, pred=pred, dropout=
         dropout, out=datasets.bmi_mono_new);

data datasets.bmi_mono_new;
    set datasets.bmi_mono_new;

    if Bmi <= 25 then BMICAT = 1;
    else if 25 < Bmi <= 30 then BMICAT = 2;
    else if Bmi > 30 then BMICAT = 3;

    timecont = time;
/*    sexcont = sex; */
run;
data datasets.bmi_mono_new;
set datasets.bmi_mono_new;
if baseline_smoker <= 0 then
baseline_smoker_new = 0;
else
baseline_smoker_new = 1;
run;
proc sort data=datasets.bmi_mono_new;
    by _imputation_;
run;
proc genmod data=datasets.bmi_mono_new;
by _imputation_;
scwgt wi;
class id time sex baseline_smoker_new BMICAT;
model BMICAT = timecont fage sex baseline_smoker_new timecont*
          baseline_smoker_new timecont*fage timecont*sex/
```

```
dist = multinomial link=cumlogit type3;
repeated subject=id/ type=ind covb corrw
within=time*sex modelse ecovb;
ods output GEEEmpPEst=datasets.gmparms parminfo=datasets.gmpinfo
modelinfo=datasets.modelinfo GEERCov=datasets.gmcovb;
run;
/*Remove the reference parameter*/
data datasets.gmpinfo;
  set datasets.gmpinfo;

  * Check condition to delete rows;
  if 1 <= _imputation_ <= 10 and (Parameter = 'Prm4' or Parameter
    = 'Prm6' or Parameter = 'Prm8' or Parameter = 'Prm11') then
    delete;
run;

/*Remove the reference group*/
data datasets.gmparms;
  set datasets.gmparms;

  * Check condition to delete rows;
  if 1 <= _imputation_ <= 10 and ( Level1 = '1') then delete;
run;

proc mianalyze parms=datasets.gmparms parminfo=datasets.gmpinfo
  covb=datasets.gmcovb wcov bcov tcov;
modeleffects intercept1 intercept2 timecont fage sex baseline_
  smoker_new timecont*baseline_sm timecont*fage timecont*sex;
run;

/*GEE MI*/
proc sort data=datasets.bmi_impute;
  by _imputation_ id fage baseline_smoker sex;
run;
proc sort data=datasets.bmi_impute;
  by sex;
run;
PROC MI DATA=datasets.bmi_impute SEED=54321 OUT=datasets.bmi_
  impute
SIMPLE NIMPUTE=1 ROUND=0.1 MINIMUM=0;
VAR fage baseline_smoker t0 t1 t2 t3 t4 t5 t6 t7 t8 t9;
BY sex; RUN;

/*Transpose from wide to long format*/
DATA datasets.bmi_longMI;
SET datasets.bmi_impute;
```

```
time=0;Bmi=t0;OUTPUT;
time=1;Bmi=t1;OUTPUT;
time=2;Bmi=t2;OUTPUT;
time=3;Bmi=t3;OUTPUT;
time=4;Bmi=t4;OUTPUT;
time=5;Bmi=t5;OUTPUT;
time=6;Bmi=t6;OUTPUT;
time=7;Bmi=t7;OUTPUT;
time=8;Bmi=t8;OUTPUT;
time=9;Bmi=t9;OUTPUT;
DROP t0-t9;
RUN;

data datasets.bmi_longMI;
    set datasets.bmi_longMI;

    if Bmi <= 25 then BMICAT = 1;
    else if 25 < Bmi <= 30 then BMICAT = 2;
    else if Bmi > 30 then BMICAT = 3;

    timecont = time;
run;
data datasets.bmi_longMI;
set datasets.bmi_longMI;
if baseline_smoker <= 0 then
baseline_smoker_new = 0;
else
baseline_smoker_new = 1;
run;
proc sort data=datasets.bmi_longMI;
    by _imputation_;
run;

proc gee data=datasets.bmi_longMI;
class id time sex baseline_smoker_new;
by _Imputation_;
model BMICAT = timecont fage sex baseline_smoker_new timecont*
    baseline_smoker_new timecont*fage timecont*sex/
dist = multinomial link=cumlogit type3;
repeated subject=id/ type=ind covb corrw
within=time(sex) modelse ecovb;
ods output GEEEmpPEst=datasets.gmparms1 parminfo=datasets.gmpinfo1
modelinfo=datasets.modelinfo1 GEERCov=datasets.gmcovb1;
run;

/*Remove the reference parameter*/
```



```
data datasets.gmpinfo1;
  set datasets.gmpinfo1;

  * Check condition to delete rows;
  if 1 <= _imputation_ <= 10 and (Parameter = 'Prm4' or Parameter
    = 'Prm6' or Parameter = 'Prm8' or Parameter = 'Prm11') then
    delete;
run;
/*Remove the reference group*/
data datasets.gmparms1;
  set datasets.gmparms1;

  * Check condition to delete rows;
  if 1 <= _imputation_ <= 10 and ( Level1 = '1') then delete;
run;

proc mianalyze parms=datasets.gmparms1 parminfo=datasets.gmpinfo1
  covb=datasets.gmcovb1 wcov bcov tcov;
modeleffects intercept1 intercept2 timecont fage sex baseline_
  smoker_new timecont*baseline_sm timecont*fage timecont*sex;
run;

/*Sensitivity Analysis*/
/*MNAR ACMV | CCMV | NCMV */

/*Convert to Monotone missingness*/
proc mi data=datasets.Bmilda_wide seed=54321 simple nimpute=10
  round= 0.1 out=datasets.bmi_MNAR;
title 'Monotone multiple imputation';
by sex;
var fage baseline_smoker t0 t1 t2 t3 t4 t5 t6 t7 t8 t9;
mcmc impute = monotone;
EM maxiter = 10000;
run;
proc sort data=datasets.bmi_MNAR;
  by _imputation_ id fage baseline_smoker sex;
run;
proc sort data=datasets.bmi_MNAR;
  by sex;
run;
/*----- CCMV*/
proc mi data=datasets.bmi_MNAR seed=54321 simple out=datasets.bmi_
  CCMV nimpute=1;
EM MAXITER=10000;
var sex fage baseline_smoker t0 t1 t2 t3 t4 t5 t6 t7 t8 t9;
```

```
monotone reg;
mnar model (t0 t1 t2 t3 t4 t5 t6 t7 t8 t9 / modelobs=ccmv);
run;
proc sort data=datasets.bmi_CCMV;
  by _imputation_ id fage baseline_smoker sex;
run;
/*Long transform*/
DATA datasets.bmi_lng_CCMV;
SET datasets.bmi_CCMV;
time=0; Bmi=t0; OUTPUT;
time=1; Bmi=t1; OUTPUT;
time=2; Bmi=t2; OUTPUT;
time=3; Bmi=t3; OUTPUT;
time=4; Bmi=t4; OUTPUT;
time=5; Bmi=t5; OUTPUT;
time=6; Bmi=t6; OUTPUT;
time=7; Bmi=t7; OUTPUT;
time=8; Bmi=t8; OUTPUT;
time=9; Bmi=t9; OUTPUT;
DROP t0-t9;
RUN;
data datasets.bmi_lng_CCMV;
  set datasets.bmi_lng_CCMV;

  if Bmi <= 25 then BMICAT = 1;
  else if 25 < Bmi <= 30 then BMICAT = 2;
  else if Bmi > 30 then BMICAT = 3;

  timecont = time;
run;
data datasets.bmi_lng_CCMV;
set datasets.bmi_lng_CCMV;
if baseline_smoker <= 0 then
baseline_smoker_new = 0;
else
baseline_smoker_new = 1;
run;
proc mixed data=datasets.bmi_lng_CCMV method=REML;
class id time sex baseline_smoker_new;
by _Imputation_;
model Bmi = timecont fage sex baseline_smoker_new timecont*
  baseline_smoker_new timecont*fage timecont*sex/ covb s;
repeated time / type = sp(exp)(timecont) local subject=id;
random intercept / subject=id;
random intercept timecont / type=un subject=sex(id);
ods output solutionF = datasets.nlparms CovB=datasets.nlcovb;
```

```
run;

/*Convert to Monotone missingness*/
proc mi data=datasets.Bmilda_wide seed=54321 simple nimpute=10
    round= 0.1 out=datasets.bmi_MNAR;
title 'Monotone multiple imputation';
by sex;
var fage baseline_smoker t0 t1 t2 t3 t4 t5 t6 t7 t8 t9;
mcmc impute = monotone;
EM maxiter = 10000;
run;
proc sort data=datasets.bmi_MNAR;
    by _imputation_ id fage baseline_smoker sex;
run;
proc sort data=datasets.bmi_MNAR;
    by sex;
run;
/*----- NCMV*/
proc mi data=datasets.bmi_MNAR seed=54321 simple out=datasets.bmi_
    NCMV nimpute=1;
EM MAXITER=10000;
var sex fage baseline_smoker t0 t1 t2 t3 t4 t5 t6 t7 t8 t9;
monotone reg;
mnar model (t0 t1 t2 t3 t4 t5 t6 t7 t8 t9 / modelobs=ncmv);
run;
proc sort data=datasets.bmi_NCMV;
    by _imputation_ id fage baseline_smoker sex;
run;
/*Long transform*/
DATA datasets.bmi_lng_NCMV;
SET datasets.bmi_NCMV;
time=0;Bmi=t0;OUTPUT;
time=1;Bmi=t1;OUTPUT;
time=2;Bmi=t2;OUTPUT;
time=3;Bmi=t3;OUTPUT;
time=4;Bmi=t4;OUTPUT;
time=5;Bmi=t5;OUTPUT;
time=6;Bmi=t6;OUTPUT;
time=7;Bmi=t7;OUTPUT;
time=8;Bmi=t8;OUTPUT;
time=9;Bmi=t9;OUTPUT;
DROP t0-t9;
RUN;
data datasets.bmi_lng_NCMV;
    set datasets.bmi_lng_NCMV;
```

```
        if Bmi <= 25 then BMICAT = 1;
        else if 25 < Bmi <= 30 then BMICAT = 2;
        else if Bmi > 30 then BMICAT = 3;

        timecont = time;
run;
data datasets.bmi_lng_NCMV;
set datasets.bmi_lng_NCMV;
if baseline_smoker <= 0 then
baseline_smoker_new = 0;
else
baseline_smoker_new = 1;
run;
proc mixed data=datasets.bmi_lng_NCMV method=REML;
class id time sex baseline_smoker_new;
by _Imputation_;
model Bmi = timecont fage sex baseline_smoker_new timecont*
        baseline_smoker_new timecont*fage timecont*sex/ covb s;
repeated time / type = sp(exp)(timecont) local subject=id;
random intercept / subject=id;
random intercept timecont / type=un subject=sex(id);
ods output solutionF = datasets.nlparms1 CovB=datasets.nlcovb1;
run;

proc mianalyze parms=datasets.nlparms11 covb(effectvar=rowcol)=
        datasets.nlcovb11;
modeleffects intercept timecont fage sex baseline_smoker_new
        timecont*baseline_sm timecont*fage timecont*sex;
run;

/*Convert to Monotone missingness Complete MI*/
proc mi data=datasets.Bmilda_wide seed=54321 simple nimpute=10
        round= 0.1 out=datasets.bmi_MNAR;
title 'Monotone_multiple_imputation';
by sex;
var fage baseline_smoker t0 t1 t2 t3 t4 t5 t6 t7 t8 t9;
mcmc impute = monotone;
EM maxiter = 10000;
run;
proc sort data=datasets.bmi_MNAR;
        by _imputation_ id fage baseline_smoker sex;
run;
proc sort data=datasets.bmi_MNAR;
        by sex;
run;
/*----- ACMV*/
```

```
PROC MI DATA=datasets.bmi_MNAR SEED=54321 OUT=datasets.bmi_ACMV
SIMPLE NIMPUTE=1 ROUND=0.1 MINIMUM=0;
VAR fage baseline_smoker t0 t1 t2 t3 t4 t5 t6 t7 t8 t9;
BY sex; RUN;
proc sort data=datasets.bmi_ACMV;
  by _imputation_ id fage baseline_smoker sex;
run;
/*Long transform*/
DATA datasets.bmi_lng_ACMV;
SET datasets.bmi_ACMV;
time=0;Bmi=t0;OUTPUT;
time=1;Bmi=t1;OUTPUT;
time=2;Bmi=t2;OUTPUT;
time=3;Bmi=t3;OUTPUT;
time=4;Bmi=t4;OUTPUT;
time=5;Bmi=t5;OUTPUT;
time=6;Bmi=t6;OUTPUT;
time=7;Bmi=t7;OUTPUT;
time=8;Bmi=t8;OUTPUT;
time=9;Bmi=t9;OUTPUT;
DROP t0-t9;
RUN;
data datasets.bmi_lng_ACMV;
  set datasets.bmi_lng_ACMV;

  if Bmi <= 25 then BMICAT = 1;
  else if 25 < Bmi <= 30 then BMICAT = 2;
  else if Bmi > 30 then BMICAT = 3;

  timecont = time;
run;
data datasets.bmi_lng_ACMV;
set datasets.bmi_lng_ACMV;
if baseline_smoker <= 0 then
baseline_smoker_new = 0;
else
baseline_smoker_new = 1;
run;
proc mixed data=datasets.bmi_lng_ACMV method=REML;
class id time sex baseline_smoker_new;
by _Imputation_;
model Bmi = timecont fage sex baseline_smoker_new timecont*
  baseline_smoker_new timecont*fage timecont*sex/ covb s;
repeated time / type = sp(exp)(timecont) local subject=id;
random intercept / subject=id;
random intercept timecont / type=un subject=sex(id);
```

```
ods output solutionF = datasets.nlparms2 CovB=datasets.nlcovb2;
run;

proc mianalyze parms=datasets.nlparms_renamed22 covb(effectvar=
    rowcol)=datasets.nlcovb_renamed22;
modeleffects intercept timecont fage sex baseline_smoker_new
    timecont*baseline_sm timecont*fage timecont*sex;
run;
```